

Setup. UCCSD doubles are compiled in the Jordan–Wigner (JW) encoding on an even register of N qubits with two spin rows: α on $0 \dots N/2-1$, β on $N/2 \dots N-1$. Each spin-paired double has a Pauli representative whose non-identity letters are Y/X on the excited orbitals plus a string of Z s (JW parity tails). Those Z s always appear as *spin pairs*: whenever Z_q occurs with $q < N/2$, so does $Z_{q+N/2}$.

Identity (exact on the reachable subspace). Starting from Hartree–Fock and applying only spin-paired doubles keeps α and β occupations locked: qubits q and $q+N/2$ always have equal occupation. Hence

$$Z_q Z_{q+N/2} = (+1) \cdot (+1) \text{ or } (-1) \cdot (-1) = +1$$

on every state the ansatz can reach. Replacing each such pair by the scalar $+1$ is therefore exact—not an approximation. In code this is deletion of both letters:

$$\underbrace{\dots Z_q \dots Z_{q+N/2} \dots}_{\text{JW string}} \longrightarrow \underbrace{\dots I \dots I \dots}_{\text{frozen string}}.$$

`freeze.freeze_z_pairs(string, keep_pairs)` performs this rewrite; `fully_freeze` deletes every freezeable pair (`keep_pairs=∅`).

Why freeze helps. Fewer non-identity Paulis $\Rightarrow$ shorter within-row CZ fan-in ladders when compiling $\exp(-i\theta P/2)$. Under the exploring error-budget proxy (cross-chip 2q cost $\gg$ on-chip), deleting idle parity tails is a pure win. Equivalence is checked by statevector overlap vs. the unfrozen circuit on the HF bitstring (overlaps ≈ 1 in `export_winners.py`).

Selective freeze: Z s as couriers. The freeze identity is bidirectional: any deleted $Z_q Z_{q+N/2}$ may be *re-inserted* for free. Retained pairs (indices $q \in \text{keep_pairs}$) act as parity *couriers* along the allowed within-spin CZ graph (nearest-neighbour + chords $(0, 3) / (N/2, N/2+3)$), so hubs and Steiner nodes sit on legal edges. `all_keep_masks` enumerates courier subsets; Method F (`method_disjoint_rzx_all_qubits`) searches masks + disjoint α – β RZX hubs jointly.

Winners in exploring/.

Molecule	Freeze	Effect
HF 6q	<i>full</i>	$\text{YZZXZZ} \rightarrow \text{YIIXII}$, etc. $\rightarrow$ three weight-2 blocks; RZX on $(0, 3), (1, 4), (2, 5)$; no CZ .
Cl ₂ 10q	<i>selective</i>	<code>keep_pairs = {{2, 3}, {3}}</code> ; $\text{IYZZXIXZZX}, \text{YIIZXXIIZX}$ stay weight ≥ 4 ; RZX on $(2, 7), (0, 5)$ with CZ fan-in.

HF collapses to independent $Y \otimes X$ pairs because (after tapering) each double has only one freezeable α – β support pair. Untapered Cl₂ keeps shared creation orbitals on qubits 4 and 9, so even maximal freeze leaves weight-4 strings—CZ ladders remain mandatory. Couriers are kept only insofar as they enable legal, vertex-disjoint hubs under the exploring connectivity rules.

Caveats. Exactness requires (i) a spin-paired reference (HF), (ii) only paired doubles in the ansatz, (iii) action on that locked subspace. The measured Hamiltonian is unchanged; only the ansatz Paulis are rewritten. Re-verify overlaps if the doubles set changes.

API: `freeze.py` — `freeze_z_pairs`, `fully_freeze`, `all_keep_masks`. Call sites: `export_winners.py`, `methods.py`.